RESUME ANALYSIS USING NLP MODELS

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CERTIFICATE

This is to certify that the thesis report entitled **"RESUME ANALYSIS USING NLP MODELS"** submitted by **Katla Lakshmi prasanna (Y22EC077), Minnakanti Revathi (Y22EC109), Lakkam Venkata Abhilash Reddy (Y22EC097),Kotte Shasi Vardhan Satakarni (Y22EC086)** in partial fulfillment for the award of the Degree of **Bachelor of Technology** in **Electronics & Communication Engineering** to R.V.R & J.C College of Engineering (Autonomous) is a record of bonafide work carried out by them under my guidance and supervision. The results embodied in this project report have not been submitted to any other University or Institute for the award of any degree or diploma.

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Abstract

Recruiters have a hard time quickly finding good candidates because manually screening resumes takes a lot of time, is not always accurate, and is subject to bias. Current Applicant Tracking Systems (ATS) depend too much on keyword matching and don't do a good job of capturing semantic relevance and understanding context.

This paper proposes an AI-based resume analysis and Recommendation System that combines Natural Language Processing (NLP), Named Entity Recognition (NER), Applicant Tracking System (ATS) scoring, and Retrieval Augmented Generation (RAG) to overcome these limitations.

Thesuggested system automates the parsing of resumes, pulls out important information like skills, education, and experience, and decides if a candidate is a good fit for the job based on the requirements. It also gives personalized feedback, analyzes skill gaps, and makes up mock interview questions to help candidates get ready. The system makes hiring easier, cuts down on bias, and lets you make decisions based on data. The results of the experiment show that this method is more accurate and better at matching meanings than traditional methods.

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1

INTRODUCTION

1.1 Introduction

In the present era, the world is witnessing rapid advancements in technology driven by continuous research and innovation across multiple domains such as Information Technology, Artificial Intelligence, Machine Learning, healthcare, and data science. These developments are transforming industries, improving efficiency, and creating a wide range of employment opportunities globally. As organizations expand and evolve, the demand for skilled professionals has increased significantly, leading to a substantial rise in the number of job applicants for various roles. [1]

In this highly competitive environment, recruitment has become a critical function for organizations. The success of any organization largely depends on its ability to hire the right candidates who possess the required skills, knowledge, and experience. The recruitment process typically begins with candidates preparing resumes that provide detailed information about their educational background, technical skills, work experience, certifications, and achievements. These resumes act as the first point of interaction between the candidate and the employer.

With the growth of digital platforms, most organizations now prefer to receive resumes through emails or online job portals. This has made the application process easier and more accessible for candidates across different regions. However, it has also resulted in an overwhelming number of applications for a single job position. Recruiters are often required to review hundreds or even thousands of resumes for a single vacancy, making the screening process extremely time consuming and inefficient. [2]

Traditionally, resume screening is performed manually by human recruiters. This involves reading each resume carefully, identifying relevant information, and comparing it with the job requirements. While this method allows for human judgment, it suffers from several limitations. One of the major drawbacks is the significant amount of time required to process large volumes of resumes. Additionally, human fatigue and workload can lead to errors, causing recruiters to overlook important details or miss highly qualified candidates.

Another critical issue associated with manual screening is bias. Recruiters may consciously or unconsciously favor certain candidates based on factors that are not directly related to job performance. This can lead to unfair hiring practices and reduce diversity within the organization. Moreover, the lack of consistency in manual evaluation can result in variations.

In order to address these challenges, there is a growing need for an automated and intelligent system that can efficiently handle the resume screening process. This is where Machine Learning (ML), a subset of Artificial Intelligence (AI), plays a vital role. Machine learning enables systems to learn from data, identify patterns, and make decisions with minimal human intervention. By leveraging ML techniques, it is possible to automate repetitive tasks, improve accuracy, and enhance the overall efficiency of recruitment processes.

A machine learning-based resume screening system is designed to analyze and process resumes in a structured manner. It uses techniques such as Natural Language Processing (NLP) to extract meaningful information from unstructured text data present in resumes. The system can identify key attributes such as skills, qualifications, experience, and keywords, and compare them with the job description to determine the suitability of candidates. Based on this analysis, candidates can be ranked or scored, enabling recruiters to focus on the most relevant applicants. Furthermore, such systems can continuously improve their performance by learning from historical hiring data. For example, by analyzing past recruitment decisions and successful employee profiles, the system can identify patterns that indicate a good match for a particular role. This makes the system more intelligent and adaptive over time. In addition, machine learning models can be trained to minimize bias by focusing only on relevant features, thereby promoting fair and unbiased hiring practices. The implementation of an automated resume screening system offers several advantages. It significantly reduces the time and effort required for initial candidate screening, allowing recruiters to concentrate on more strategic aspects of hiring such as interviews and decision-making. It also improves the consistency and accuracy of the evaluation process, ensuring that all candidates are assessed based on the same criteria. Moreover, it enhances the over-

all candidate experience by enabling faster response times and more transparent selection processes. [3, 4]

Despite its advantages, the adoption of machine learning in recruitment also presents certain challenges. These include the need for high-quality training data, the risk of algorithmic bias if not properly designed, and the requirement for continuous monitoring and updating of models. However, with proper implementation and ethical considerations, these challenges can be effectively managed.

In conclusion, the integration of machine learning into the recruitment process represents a significant advancement in modern hiring practices. As organizations continue to receive large volumes of applications, the need for efficient and intelligent screening systems becomes increasingly important. A machine learning-based resume screening system not only streamlines the recruitment process but also improves the quality of hiring decisions by identifying the most suitable candidates in a fair and efficient manner. This makes it an essential tool for organizations aiming.

1.2 Problem statement

Today's digital era, organizations across the globe are experiencing a significant increase in the number of job applications due to the rapid growth of online job portals, email-based applications, and professional networking platforms. While this has made job opportunities more accessible to candidates, it has also created a major challenge for recruiters in efficiently managing and processing large volumes of resumes.

The traditional recruitment process primarily relies on manual screening of resumes, where recruiters carefully read and evaluate each application to determine whether a candidate meets the job requirements. This process becomes increasingly difficult and inefficient when the number of applicants is very high. Recruiters are often required to review hundreds or even thousands of resumes for a single job opening, which leads to excessive time consumption and delays in the hiring process. [2, 4]

One of the major problems associated with manual resume screening is the lack of efficiency. Human recruiters have limited time and capacity, which makes it difficult to thoroughly analyze every resume. As a result, important details such as relevant skills, experience, or achievements may be overlooked. This can lead to the rejection of highly qualified candidates who may be a perfect fit for the organization.

Another significant issue is the high probability of human error. Factors such as fatigue, workload pressure, and repetitive tasks can negatively impact the recruiter's performance. This increases the chances of incorrect evaluation, inconsistent decision-making, and unintentional mistakes during the screening process. Consequently, the overall quality of candidate selection may be compromised.

In addition to inefficiency and errors, bias in recruitment is another critical concern. Manual screening may introduce conscious or unconscious bias based on factors such as educational background, gender, location, or personal preferences. This can result in unfair hiring practices and may prevent organizations from selecting the most suitable candidates purely based on merit.

Furthermore, the lack of standardization in the evaluation process makes it difficult to ensure consistency across different recruiters or hiring cycles. Different recruiters may interpret the same resume differently, leading to variations in candidate selection. This inconsistency reduces the reliability and transparency of the recruitment process.

Another major challenge is the inability to handle unstructured data effectively. Resumes are usually submitted in various formats such as PDF, Word documents, or plain text, and they contain unstructured information. Extracting meaningful insights from such data manually is both time-consuming and inefficient. Recruiters may struggle to quickly identify key information such as technical skills, years of experience, or domain expertise. [1]

Additionally, the delay in the hiring process caused by manual screening can negatively impact both organizations and candidates. Organizations may lose potential talent to competitors due to slow response times, while candidates may become frustrated due to a lack

of timely communication. This can harm the organization's reputation and candidate experience.

The growing volume of applications also makes it difficult to scale the recruitment process using traditional methods. As organizations expand, the need for a faster and more efficient screening system becomes essential. Without automation, it becomes nearly impossible to maintain speed, accuracy, and fairness simultaneously.

To address these challenges, there is a strong need for an intelligent, automated, and scalable system that can efficiently process large volumes of resumes, extract relevant information, and accurately match candidates with job requirements. Such a system should be capable of reducing manual effort, minimizing errors, eliminating bias, and improving the overall efficiency and effectiveness of the recruitment process.

Therefore, the problem lies in designing and implementing a system that leverages advanced technologies such as Machine Learning and Natural Language Processing to automate resume screening. The system should be able to analyze unstructured resume data, identify key attributes, rank candidates based on suitability, and assist recruiters in making informed hiring decisions. Solving this problem is essential for modern organizations to streamline their recruitment process and ensure that the best candidates are selected in a fair, accurate, and timely manner.

1.3 Objectives

1.3.1 Primary Objective

- The primary objective of our project is to design and develop an advanced, intelligent recruitment system that automates the entire initial hiring pipeline by integrating Machine Learning (ML) and Natural Language Processing (NLP) techniques. The system aims to efficiently screen and rank candidate resumes based on job requirements and further enhance the recruitment process by generating personalized mock interview questions for shortlisted candidates. This approach not only improves the accuracy and speed of candidate selection but also assists candidates in prepar-

ing effectively for interviews, thereby creating a comprehensive and intelligent hiring solution for modern organizations.

1.3.2 Secondary Objective

- To design and implement a fully automated resume screening system capable of processing large volumes of resumes with high efficiency and accuracy.
- To extract structured information from unstructured resume data using Natural Language Processing techniques such as tokenization, stop-word removal, stemming, and entity recognition.
- To identify and analyze key candidate attributes including technical skills, educational qualifications, certifications, projects, and work experience.
- To develop a robust candidate-job matching algorithm that compares resume content with job descriptions and calculates similarity scores.
- To design a ranking system that prioritizes candidates based on relevance, experience, and skill compatibility with the job role.
- To reduce manual effort and eliminate repetitive tasks performed by human resource teams during the initial screening process.
- To minimize human errors and inconsistencies in resume evaluation and candidate shortlisting.
- To ensure fairness and reduce bias in recruitment by relying on data-driven and objective evaluation methods.

1.4 Scope of the Project

The scope of this project focuses on the design and development of an intelligent, automated recruitment system that streamlines the initial stages of the hiring process. The system integrates Machine Learning (ML) and Natural Language Processing (NLP) techniques to

efficiently process resumes, identify suitable candidates, and generate personalized mock interview questions, thereby enhancing both recruiter efficiency and candidate preparedness.

1.4.1 Covered Features

- **Resume Data Collection and Processing:** The system is designed to accept resumes in multiple formats such as PDF, Word documents, and plain text files. It processes these documents using Natural Language Processing (NLP) techniques to extract relevant information such as skills, education, work experience, certifications, and other important details required for evaluation.

- **Automated Resume Screening:** The project focuses on automating the manual resume screening process. Instead of recruiters reviewing each resume individually, the system analyzes all resumes efficiently and filters candidates based on predefined job requirements, thereby reducing time and effort.

- **Candidate-Job Matching:** The system compares extracted resume data with job descriptions to determine the suitability of each candidate. It evaluates parameters such as skill match, experience level, and domain knowledge to ensure accurate matching between candidates and job roles.

- **Candidate Ranking System:** A ranking mechanism is implemented to assign scores to candidates based on their relevance to the job. The system generates a prioritized list of applicants, helping recruiters quickly identify the most suitable candidates without manual effort.

- **Mock Interview Question Generation:** One of the key features within the scope of this project is the generation of personalized mock interview questions for shortlisted candidates.The system generates questions based on:

- Candidate's technical skills

- Experience level

- Job role requirements

These questions are categorized into:

- Technical questions
- Behavioral questions
- HR questions

This helps candidates prepare effectively for interviews.

2

LITERATURE SURVEY

2.1 Introduction

In today's competitive job market, organizations receive a large number of resumes for every job opening, making manual screening a time-consuming and inefficient process. Traditional recruitment methods often rely on keyword matching and human judgment, which can lead to inconsistencies, bias, and missed opportunities to identify suitable candidates. With the rapid growth of Natural Language Processing (NLP), automated resume analysis systems have emerged as an effective solution to streamline recruitment. These systems are capable of extracting relevant information such as skills, education, and experience, and matching it with job descriptions to identify the most suitable candidates. By leveraging advanced NLP techniques and machine learning models, resume analysis systems not only improve efficiency but also enhance accuracy and fairness in the hiring process. [5, 6]

2.2 Literature Survey

Several research studies have explored the use of NLP and machine learning techniques for automated resume analysis. Early systems relied on rule-based and keyword-matching approaches, which lacked the ability to understand context and semantics. To overcome these limitations, machine learning models such as Support Vector Machines and Decision Trees were introduced, improving classification accuracy but still requiring manual feature extraction. With the advancement of deep learning, techniques like Named Entity Recognition (NER) have been widely used to extract structured data such as names, skills, and qualifications from resumes using tools like spaCy and NLTK. Furthermore, transformer-based models such as BERT have significantly improved semantic understanding, enabling systems to analyze the meaning of text rather than relying solely on keywords. Recent research also incorporates Retrieval-Augmented Generation (RAG) models to provide explanations, feedback, and better matching between resumes and job descriptions. Despite these advancements, existing systems still face challenges such as bias, lack of explainability, and difficulty handling diverse resume formats.[6]

Several research studies have explored the use of Natural Language Processing (NLP)

and machine learning techniques for automated resume analysis. Early systems primarily relied on rule-based methods and keyword matching, which were simple to implement but lacked the ability to understand contextual meaning, resulting in poor generalization and inaccurate candidate screening. These systems often failed when synonyms or varied phrasing were used, limiting their effectiveness in real-world scenarios.[7,8]

To address these limitations, traditional machine learning models such as Support Vector Machines (SVM), Naïve Bayes, and Decision Trees were introduced. These approaches improved classification performance by learning patterns from data; however, they still depended heavily on manual feature engineering, such as extracting keywords, term frequency–inverse document frequency (TF-IDF), and syntactic features. This process was time-consuming and required domain expertise, making the systems less scalable and adaptable.[8]

S.NO	Author(year)	Title	Methodology	Results
1.	Pratik G. Raut, Prof. Rajesh D. Wagh(2024)	Resume Analyzer and Recommender System Using Python	Built with Python, Pyresparser, MySQL, and Streamlit to parse, score, and rank resumes securely with cloud deployment and GDPR compliance.	Achieved automated resume parsing, scoring, and ranking with secure data handling and intuitive visualization, enabling recruiters to quickly identify top candidates.
2.	Arvind Kumar Sinha, Md Amir Khusru Akhtar, Mohit Kumar(2023)	Automated Resume Parsing and Job Domain Prediction using Machine Learning.	Used NLP preprocessing (text extraction, cleaning, tokenization, lemmatization), entity recognition (spaCy, regex), and machine learning (Random Forest) to parse resumes and predict job domains.	Achieved 92.08% accuracy and 0.92 F1-score, with strong precision and recall across IT, Healthcare, Engineering, Marketing, Finance, and Education domains.
3.	G. Krishna Lohith, K. Varshith Reddy, P. Mounisha Reddy(2025)	NLP-Based Resume Analysis, Skill Enhancement & Job Automation.	Used NLP and ML for resume parsing, skill gap detection, course recommendations, and job matching via cosine similarity.	Achieved effective resume analysis, personalized learning paths, AI-based career guidance, and integrated job search support.
4.	K. L. Abhishek, M. Niranjanamurthy(2025)	Developing an Intelligent Resume Screening Tool With AI-Driven Analysis and Recommendation Features.	Built an AI-powered resume analyzer using Python with NLP libraries (Pyresparser, NLTK), Streamlit for interface, and MySQL for structured storage. The tool extracts key details, provides skill recommendations, feedback, and analytics for recruiters and applicants.	Achieved improved resume parsing, skill recommendations, personalized feedback, analytics, and higher accuracy (85%) compared to existing tools like Hloom (80%) and ZipRecruiter (78%).

Figure 2.1: Literature survey

2.3 Summary

In summary, the evolution of resume analysis systems has progressed from simple keyword-based methods to advanced NLP and deep learning models capable of understanding context and semantics. While modern approaches using techniques like NER, transformer models, and RAG have significantly improved accuracy and efficiency, there remain gaps in terms of transparency, fairness, and adaptability. The proposed system aims to address these limitations by integrating multiple techniques, including resume parsing, semantic matching, and feedback generation, to create a more robust and intelligent recruitment solution. This approach not only enhances candidate-job matching but also provides meaningful insights and guidance to applicants, ultimately improving the overall recruitment process, while also ensuring scalability for handling large volumes of resumes and enabling continuous learning to adapt to evolving job market trends. [6, 7]

Title	Approach / Techniques	Tools / Technologies	Outcome
Resume Parsing & Domain Prediction (2023)	ML-based parsing & job domain prediction	NLP, regex, spaCy, Random Forest	Achieved 80% accuracy
Resume Analyzer & Recommender (2024)	Python-based parsing & recommender with security	Pyresparser, MySQL, Streamlit, cloud deployment	Achieved efficient, secure & scalable analysis
Intelligent Screening Tool (2025)	AI-driven parsing with feedback & recommendations	Pyresparser, NLTK, MySQL, Streamlit	Achieved 85% accuracy (better than existing tools)
NLP-Based Resume Analysis (2025)	NLP + ML for parsing & skill gap detection	NLP, cosine similarity, course suggestions	Achieved effective skill detection & AI support

Figure 2.2: Comparsion Table

The three papers illustrate the evolution of resume analysis systems from basic parsing and ranking (2024), to more accurate prediction using NLP and ML techniques like Random Forest (2023), and finally to advanced AI-driven systems (2025) that include skill gap detection, course recommendations, and job matching. The later approaches improve accuracy, automation, and user experience. Overall, this progression highlights a shift toward

intelligent, efficient, and personalized recruitment solutions that benefit both recruiters and job seekers.

3

THEORETICAL BACKGROUND

3.1 Introduction To Machine Learning

Machine Learning (ML) is a subset of Artificial Intelligence that enables computers to learn from data and make decisions without being explicitly programmed. Instead of relying on predefined rules, ML systems analyze large amounts of data, identify patterns, and improve their performance over time. It plays a significant role in modern technology by automating complex tasks, increasing accuracy, and efficiently handling large-scale data. Machine Learning is generally classified into three main types: supervised learning, unsupervised learning, and reinforcement learning, each designed to solve different kinds of problems such as prediction, pattern recognition, and decision-making. [3, 4]

At its core, machine learning works by training a model using historical data. The model learns patterns from this data and then applies that knowledge to new, unseen data to make predictions. The performance of a machine learning system improves as more data.

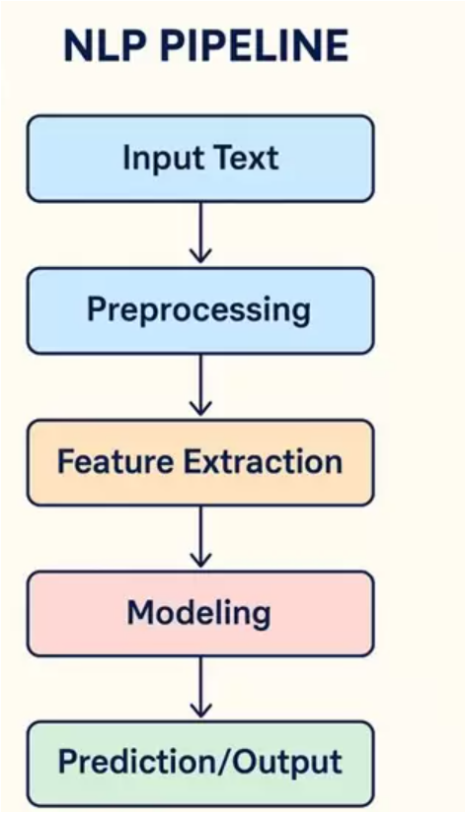


Figure 3.1: NLP Architecture

3.2 Natural Language Processing Model Pipeline

NLP models are algorithms or computational techniques used to understand, process, and generate human language. These models take textual data as input and perform tasks such as classification, translation, sentiment analysis, and information extraction. Depending on complexity, NLP models can be divided into traditional machine learning models and advanced deep learning models.

3.2.1 Input Text: The NLP pipeline begins with input text, which is raw and unstructured data collected from various sources such as resumes, emails, documents, social media posts, or web content. This data may contain inconsistencies such as different formats, spelling variations, symbols, and irrelevant information. Since machines cannot directly understand human language, this raw input must be processed through multiple stages to extract meaningful information. [6, 7]

3.2.2 Preprocessing: Preprocessing is a critical step in the NLP pipeline where the raw text is cleaned and standardized to make it suitable for analysis. The quality of preprocessing directly affects the performance of the model.
This stage includes several sub-steps:

- Tokenization: Splitting text into smaller units such as words or sentences.
- Lowercasing: Converting all text into a uniform format (e.g., lowercase).
- Stop-word Removal: Removing common words like “is”, “the”, “and” that do not add significant meaning.
- Stemming and Lemmatization: Reducing words to their root or base form.
- Removing Noise: Eliminating punctuation, special characters, numbers, and extra spaces.

3.2.3 Feature Extraction: After preprocessing, the text needs to be converted into a numerical representation because machine learning models cannot understand text directly. This process is called feature extraction.

Common techniques used:

- Bag of Words (BoW): Represents text based on word frequency.
- TF-IDF: Assigns importance to words based on their frequency across documents.
- Word Embeddings: Converts words into dense vectors capturing semantic meaning (e.g., Word2Vec, GloVe)

  This stage transforms textual data into structured data that can be fed into machine learning models.

3.2.4 Modeling: In this stage, machine learning or deep learning models are applied to the extracted features. The model learns patterns, relationships, and structures within the data during training.

Common models include:

- Machine Learning: Naive Bayes, SVM, Decision Trees.
- Deep Learning: RNN, LSTM, Transformer models like BERT

  The choice of model depends on the complexity of the task and the dataset size. The model is trained using labeled or unlabeled data depending on the problem type.

3.2.5 Prediction/Output: This is the final stage of the NLP pipeline where the trained model produces results based on the input data. The output can vary depending on the application.

Examples:

- Classification (e.g., spam detection)
- Sentiment analysis (positive/negative)

- Resume ranking (in your project)
- Question generation (mock interview questions)

  The output is meaningful and actionable, helping users make decisions or automate tasks.

3.3 Applications

Natural Language Processing (NLP) is widely used in various real-world applications to enable machines to understand, interpret, and generate human language. It plays a crucial role in improving interaction between humans and computers and is used across multiple domains.

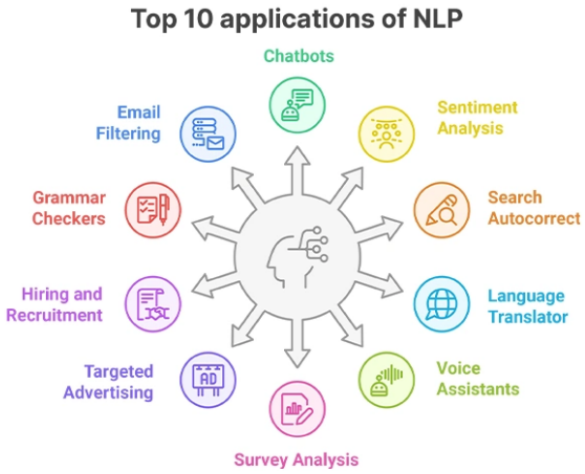


Figure 3.2: Applications of NLP

3.3.1 Chatbots: Customer service and experience are the most important things for any company. It can help companies improve their products and also keep customers satisfied. But interacting with every customer manually and resolving the problems can be a tedious task. This is where Chatbots come into the picture. Chatbots help companies achieve the goal of a smooth customer experience.

3.3.2 Sentiment Analysis: More and more people these days have started using social media to post their thoughts about a particular product, policy, or matter. These could con-

tain some useful information about an individual's likes and dislikes. Hence, analyzing this unstructured data can help in generating valuable insights. Natural Language Processing comes to the rescue here, too. It plays a crucial role in sentiment analysis, especially for customer data.

3.3.3 Search Autocorrect and Autocomplete: It's a wonderful application of natural language processing and a great example of how it is affecting millions around the world, including you and me. Search autocomplete and autocorrect both help us find accurate results more efficiently. Now, various other companies have also started using this feature on their websites, like Facebook and Quora.
The driving engine behind search-autocomplete and autocorrect is the language models.

3.3.4 Language Translator: Today, tools like Google Translate can easily convert text from one language to another. These tools are helping numerous people and businesses in breaking the language barrier and becoming successful.

3.3.5 Voice Assistants: A voice assistant is a software that uses speech recognition, natural language understanding, and natural language processing to understand the verbal commands of a user and perform actions accordingly. You might say it is similar to a chatbot, but I have included voice assistants separately because they deserve a better place on this list. They are much more than a chatbot and can do many more things than a chatbot can.

3.3.6 Survey Analysis: Surveys are an important way of evaluating a company's performance. Companies conduct many surveys to get customers' feedback on various products. This can be very useful in understanding the flaws and helping companies improve their products. But the problem arises when a lot of customers take the survey, leading to an increasing data size. It becomes impossible for a person to read them all and draw a conclusion. That's where companies use natural language processing to analyze the surveys and generate insights from them, like knowing the sentiments of users about an event from the feedback and analyzing product reviews to understand the pros and cons. Today, most

companies use these methods because they provide much more accurate and useful information.

4

SYSTEM DESIGN AND IMPLEMENTATION

4.1 Existing System

Traditional resume screening systems mainly rely on manual processes where recruiters read and evaluate resumes individually. This approach is time-consuming and inefficient, especially when organizations receive thousands of applications for a single job role. Due to time constraints, recruiters may not thoroughly evaluate each resume, leading to inconsistent hiring decisions and potential loss of talented candidates. [3, 6]

With the introduction of Applicant Tracking Systems (ATS), the process became partially automated. These systems filter resumes based on keywords, predefined rules, and formatting patterns. However, they primarily depend on exact keyword matching and lack the ability to understand context or semantics. As a result, candidates with relevant skills.

Existing systems also struggle with unstructured data, as resumes come in various formats such as PDFs, Word files, and creative templates. Many ATS tools fail to properly parse resumes with complex layouts, tables, or graphics, leading to incorrect data extraction or rejection of valid applications.

Additionally, traditional systems lack transparency and feedback mechanisms. Candidates are often not informed why their resumes were rejected, making it difficult for them to improve. The absence of intelligent analysis and explanation reduces trust in automated recruitment systems.

[Resume Upload] $\rightarrow$ [Keyword Matching] $\rightarrow$ [Basic Filtering] $\rightarrow$ [Shortlisting]

4.2 Limitations of Existing Architecture

One major limitation of existing systems is their heavy reliance on keyword matching. These systems fail to capture semantic meaning and contextual relationships between words, which can result in rejecting qualified candidates who use different terminology to describe their skills.

Another limitation is bias in decision-making. Manual screening introduces human bias, while automated systems may inherit bias from training data. This can lead to unfair can-

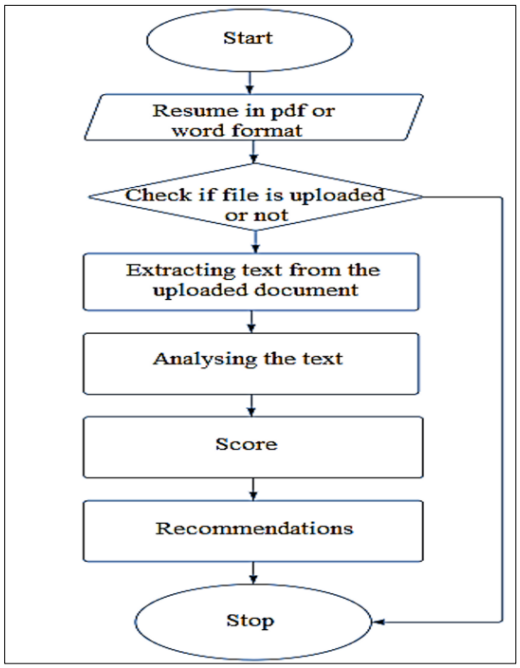


Figure 4.1: Existing system

didate selection and reduced diversity in hiring processes.

Existing systems also face challenges in handling diverse resume formats. Complex layouts, images, and non-standard structures often confuse parsing algorithms, leading to data extraction errors or loss of important information.

Most resume screening systems are trained on generic datasets and are not easily adaptable to specific industries such as healthcare, finance, or engineering. This lack of domain adaptation reduces their ability to accurately interpret specialized skills, certifications, and job roles.

Soft skills such as communication, leadership, teamwork, and problem-solving are critical in hiring decisions. However, existing systems struggle to extract and evaluate these skills accurately because they are often implied rather than explicitly mentioned. This leads to incomplete candidate evaluation.

Many systems perform well only when resumes follow a structured format. If a resume is creative or uses unconventional layouts, the system may fail to extract information correctly. This limitation reduces flexibility and may penalize candidates with well-designed but non-standard resumes.

Finally, lack of explainability is a critical issue. Most systems do not provide reasons for candidate rejection or selection, making the process opaque. This reduces user trust and prevents candidates from improving their resumes effectively. [1, 2]

4.3 Proposed System

The proposed system introduces an intelligent resume analysis framework using Natural Language Processing (NLP) and machine learning techniques. It aims to automate resume screening while understanding the semantic meaning of text, enabling more accurate candidate evaluation.

The system uses advanced techniques such as Named Entity Recognition (NER) to extract structured information like skills, education, and experience from resumes. This structured data is then compared with job descriptions for better matching.

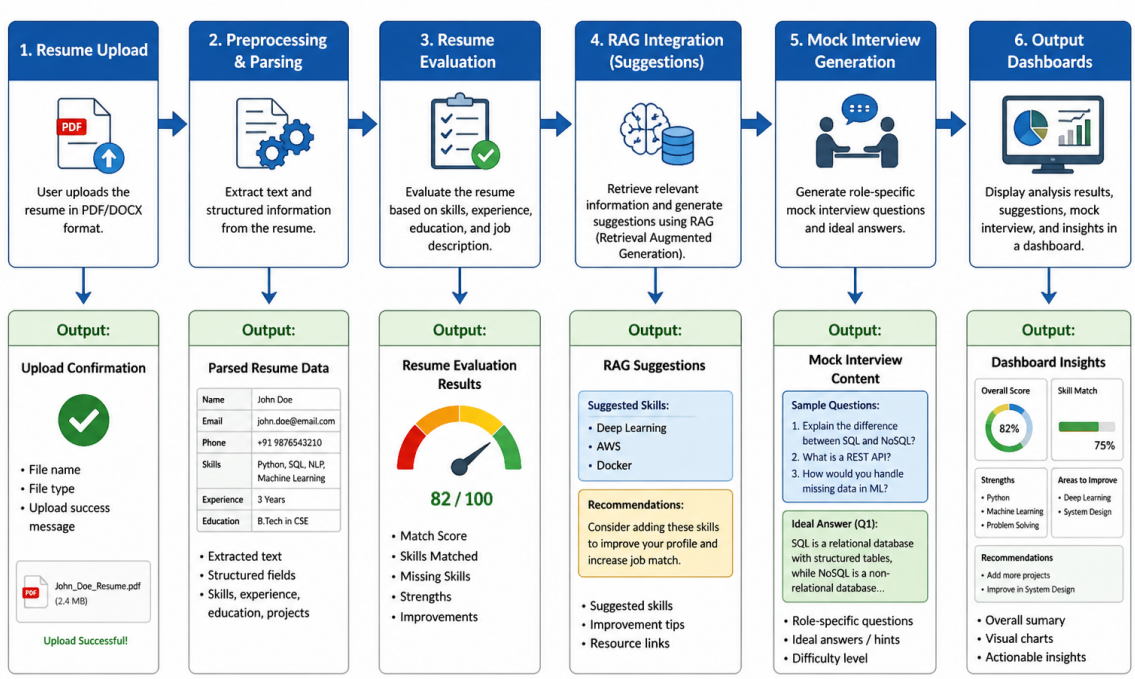


Figure 4.2: Proposed system

In addition, the system integrates semantic similarity models that analyze the meaning of both resumes and job descriptions rather than relying solely on keywords. This ensures that candidates with relevant skills are identified even if different wording is used.

The proposed system also includes a feedback and explanation module, which provides insights into why a resume matches or does not match a job. This improves transparency and helps candidates enhance their resumes.

4.4 Software Tools and Modules Used

The system uses NLP libraries such as spaCy and NLTK for text processing and entity extraction. These tools help in identifying important components like skills, organizations, and qualifications from resumes.

Machine learning and deep learning models such as BERT and other transformer-based architectures are used for semantic understanding and similarity matching. These models improve accuracy by capturing contextual meaning.

The backend of the system can be developed using Python frameworks, while the frontend can use technologies like React.js for user interaction. Databases are used to store resumes, job descriptions, and analysis results.

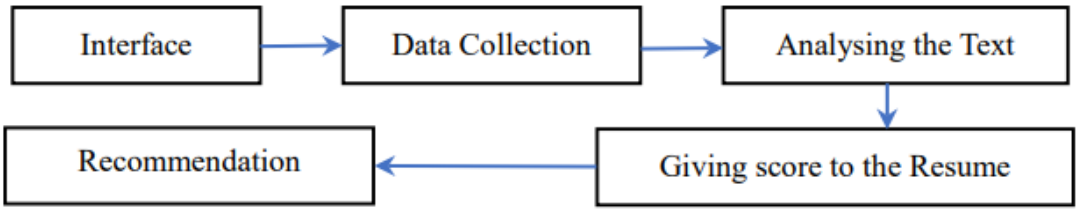


Figure 4.3: Module Diagram

The system is divided into multiple modules, including Resume Parser, NER Module, ATS Scoring Module, Semantic Matching Module, and Feedback Generation Module. Each module works together to provide a complete resume analysis solution.

1. Named Entity Recognition

- Named Entity Recognition (NER) is an NLP technique used to identify and classify important information from unstructured text.

- It detects entities such as name, skills, education, experience organization, and dates from resumes.
- NER converts raw resume text into structured data, which can be easily analyzed by the system.
- After extracting entities, the system compares resume entities with job requirements.
- Each important entity is assigned a weight (points) based on its importance. Skills → High weight, Experience → Medium weight, Education & Certifications → Lower weight
- Points are given when resume entities match job-required entities.
- The total points contribute to the overall ATS score

2. ATS & RAG Modules

- Applicant Tracking System (ATS) is used to evaluate and rank resumes based on job requirements.
- It compares extracted resume data with the job description using NLP techniques.
- ATS generates a match score that indicates how suitable a resume is for a particular role.
- Retrieval-Augmented Generation (RAG) is an advanced NLP technique that combines information retrieval with text generation.
- Instead of generating answers only from a trained model, RAG first retrieves relevant data from an external knowledge source.
- Using the retrieved information, the system then generates accurate and context-aware responses

3. Visual studio code : Visual studio code is used as the development environment for implementing, training, and testing the resume analysis model. It provides an interactive environment with GPU support for faster training.

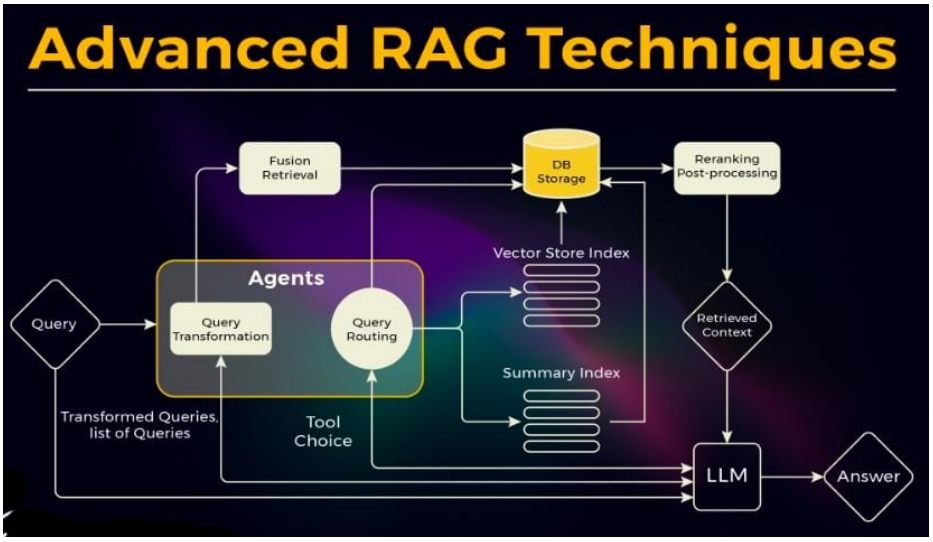


Figure 4.4: RAG Algorithm

4.5 Overall workflow

The proposed methodology aims to design and implement an intelligent, automated recruitment system that leverages Machine Learning (ML) and Natural Language Processing(NLP) techniques to streamline the resume screening process. In this system, we utilizevarious NLP models to effectively process and analyze textual data present in resumes andjob descriptions. The system is developed to efficiently handle large volumes of resumes,extract meaningful insights, evaluate candidate suitability, and generate personalized mockinterview questions for shortlisted candidates. By incorporating NLP models for tasks suchas text preprocessing, feature extraction, and semantic analysis, the system improves the ac-curacy of candidate-job matching and overall decision-making. This methodology ensures accuracy, scalability, and efficiency while reducing manual effort and improving hiring quality.

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interview questions for shortlisted candidates. By incorporating NLP models for tasks such as text preprocessing, feature extraction, and semantic analysis, the system improves the accuracy of candidate-job matching and overall decision-making. This methodology ensures accuracy, scalability, and efficiency while reducing manual effort and improving hiring quality.

The proposed system follows a structured and systematic workflow to automate the resume screening process and generate personalized mock interview questions. The workflow is designed as a multi-stage pipeline that transforms raw resume data into meaningful outputs, enabling efficient and accurate candidate selection. Each stage of the system performs a specific function, ensuring smooth data flow and effective processing.

- **Resume and Job Description Input:** The workflow begins with the input stage, where resumes are collected from candidates in various formats such as PDF, Word documents, or text files. Along with this, recruiters provide job descriptions that specify the required skills, qualifications, and experience. These inputs act as the foundation for the entire process.

- **Data Extraction and Resume Parsing:** In this stage, the system extracts relevant information from the resumes using NLP techniques. The unstructured resume data is converted into structured format by identifying key entities such as skills, education, experience, certifications, and contact details. This step ensures that important information is organized for further analysis.

- **Text Preprocessing:** The extracted text undergoes preprocessing to clean and standardize the data. This includes removing stop words, converting text to lowercase, eliminating special characters, and performing stemming or lemmatization. Preprocessing improves data quality and prepares it for feature extraction.

- **Feature Extraction:** The cleaned text is transformed into numerical representations using NLP models and techniques such as TF-IDF, Bag of Words, or word embeddings. This step allows machine learning models to understand and process the textual information effectively.

- **Candidate-Job Matching:** The system compares the features extracted from resumes with those from job descriptions. Similarity measures are used to evaluate how well each candidate matches the job requirements based on skills, experience, and keywords.

- **Candidate Scoring and Ranking:** Based on the matching results, each candidate is assigned a score. Candidates are then ranked in descending order according to their relevance and suitability for the job role. This helps recruiters focus on top candidates.

- **Shortlisting of Candidates:** Candidates who meet the required threshold are shortlisted for the next stage. This step filters out less relevant candidates and ensures that only qualified applicants proceed further.

- **Mock Interview Question Generation:** For shortlisted candidates, the system generates personalized mock interview questions. These questions are created based on the candidate's skills, experience, and job role. The questions are categorized into technical, behavioral, and HR questions to provide comprehensive interview preparation.

- **Output Generation:** The final output includes:

- Ranked list of candidates

- Candidate scores and highlights

- Generated mock interview questions

  This information is presented to recruiters in a clear and structured format.

- **User Interface and Interaction:** The system provides a user-friendly interface where recruiters can upload resumes, input job descriptions, and view results. This ensures ease of use and efficient interaction with the system.

  The overall workflow of the proposed system ensures a seamless and efficient recruitment process by integrating NLP and Machine Learning techniques. It automates re-

sume analysis, improves candidate selection accuracy, and enhances interview preparation, making it a complete and intelligent recruitment solution.

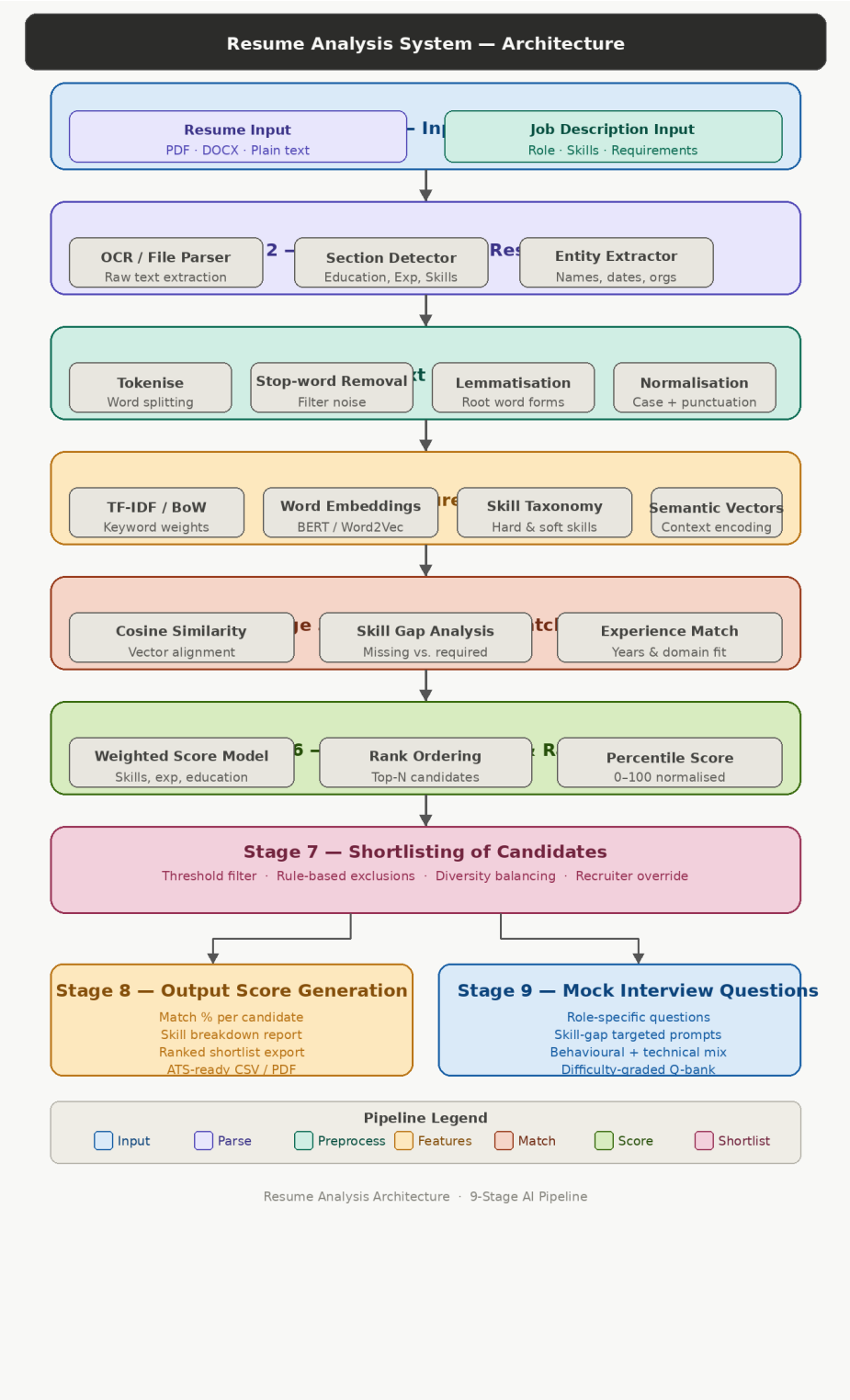


Figure 4.5: Workflow

5

RESULTS

5.1 Result

The suggested AI-based resume analysis system is tested by putting it into action and showing the results. The system shows that it works by processing resumes, getting useful information from them, and making smart suggestions.

5.1.1 Input resume 1 :

The system can read resumes that are in PDF or DOCX format. Fig 5 shows an example of a resume that was sent to the system. The input has unstructured text that includes personal information, skills, education, and work history.

5.1.2 User Interface for Resume Upload :

A user-friendly front-end interface lets users upload resumes and start the analysis process. The system can read files in PDF and DOCX formats, so you can use a variety of input types. Once the resume is uploaded, it is processed automatically without anyone having to do anything. Fig 5.3 shows the screen where users choose and upload resumes for analysis. The design makes it easy to get around and find results quickly, which improves the overall user experience.

5.1.3 Feature-Based Performance Analysis :

The system looks at different factors, like skills, education, work experience, and projects. Fig 5.6 shows these features in a graph, showing how they affect the overall score. This helps you figure out what parts of the resume are good and what parts need work.

5.1.4 Extracted Skills and Recommendations : The NLP and NER modules pull out the relevant skills from the resume. Fig 5.7 displays the extracted skills alongside suggested skills and preparation guidelines. The system finds weak or missing areas and gives suggestions for how to make them better.

5.1.5 Domain-Specific Recommendations :

The system makes recommendations based on the candi date's profile and the domain. Extracted Skills and Recommendations topics and questions for Data Science interviews. These sug gestions help candidates meet the standards of the industry.

Chandra Sekhar Chennupati

cs.chennupati@gmail.com +91 9949385751 Hyderabad, India linkedin.com/in/chandra-sekhar-chennupati

SUMMARY

I am an experienced Data Engineer skilled in PySpark, Google Cloud Platform(GCP), Python, SQL and Airflow. I specialize in building and optimizing data pipelines. I constantly commit to continuous learning and leveraging new technologies to enhance data solutions.

PROFESSIONAL EXPERIENCE

Data Engineer 07/2021 – present
Tata Consultancy Services Hyderabad, India

- Build ETL/ELT/EL batch data pipelines to move data from on-premise to Data Warehouse (**BigQuery**) and vice-versa using **Pyspark**, **SparkSQL** and **Apache Airflow**
- Iteratively design and develop a modular, reusable PySpark and Airflow based **framework** to create and orchestrate distributed data processing pipelines
- Fine-tuned and migrated oracle **stored procedures** and **packages** to Bigquery
- Optimize and performance tune big data pipelines to handle TBs of data using Apache Spark and SQL
- Perform proof of concepts(**POC**) using interactive development environments such as Jupyter Notebooks
- Debug erroneous data pipelines including undocumented legacy pipelines and troubleshoot complex data issues and provide solutions
- Create **Bash** or **Python** scripts to **automate** routine tasks

EDUCATION

V R SIDDHARTHA ENGINEERING COLLEGE 2017 – 2021
Bachelor of Technology in Electronics and Communication Engineering Vijayawada, India
CGPA: 8.61

PROJECTS

Smart AI Doc Miner
Generative AI, Flask, Python, Google Cloud Platform, pgvector, Google Gemini 1.5

- Implemented **RAG (Retrieval-Augmented Generation)** methodology to enable efficient document retrieval and analysis.
- Integrated **Google Gemini 1.5 model** for state-of-the-art natural language processing and understanding.
- Utilized **pgvector** for vector storage and similarity search, enhancing data retrieval capabilities.
- **Deployment**: Hosted the application on **Google Cloud Run**, ensuring scalability and seamless deployment.

SKILLS

Programming Languages — Python, SQL, Shell Scripting

Frameworks — Apache Spark (Pyspark), Apache Airflow

Cloud Services — Google cloud platform (GCP) - BigQuery, Google Cloud Storage(GCS), Dataproc, Pub/Sub, Cloud Composer, Google Compute Engine, Google Kubernetes Engine, IAM, Cloud SQL, Cloud Functions, Vertex AI, Cloud Run, Cloud Shell, Cloud Scheduler, Artifact Registry | **Amazon Web Services (AWS)** - S3, Lambda, DynamoDB | **Microsoft Azure** - Azure Data Factory, Azure Databricks, Azure Synapse Analytics

ETL Tools — IBM InfoSphere DataStage

Orchestration Tools — Cloud Composer, Control-M

DBMS — Teradata, postgres, oracle

CERTIFICATES

- Google Cloud Certified Professional Data Engineer
- Google Cloud Certified Associate Data Practitioner
- Google Cloud GenAI L400 Certified

AWARDS

Employee of the Month x 3

- On three different occasions, I have been awarded Employee of the Month for my individual contributions

Project Performance Improvement Award x2

- As part of a 5 member team, helped the customer save more than $150k
- As part of a 2 member team, worked to meet Government's Data Regulatory policies within a strict deadline of 4 months

INTERESTS

Cricket | Volleyball | Badminton | Reading | Photography

Figure 5.1: Resume 1

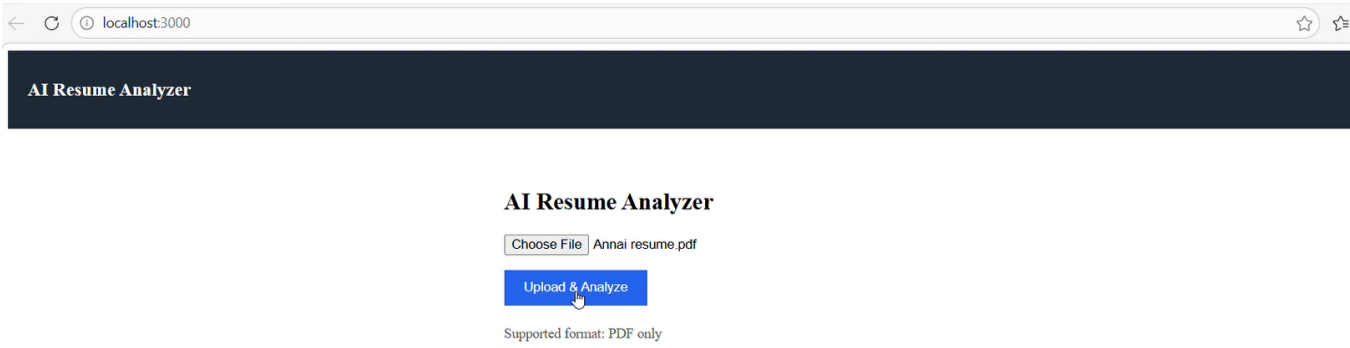


Figure 5.2: Dashboard Page

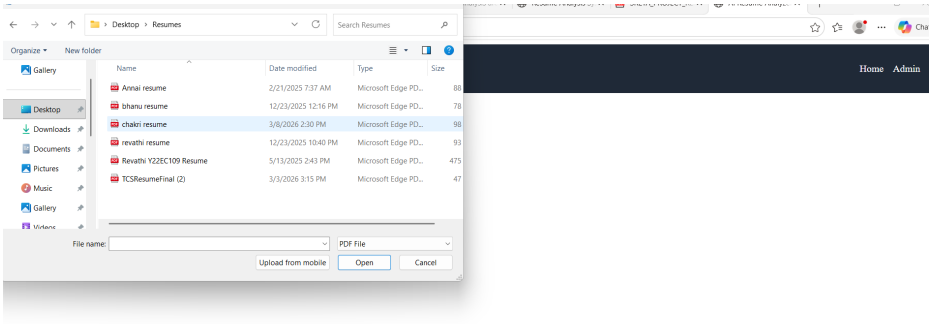


Figure 5.3: Uploading Resume

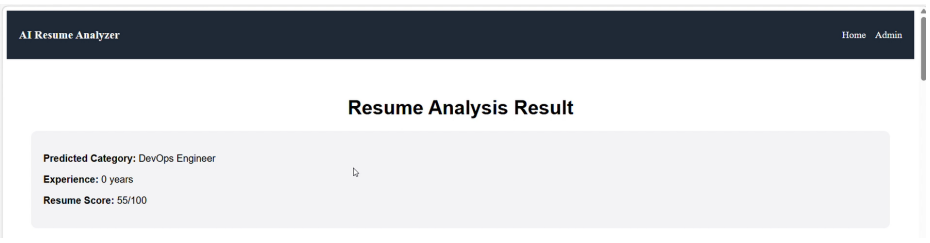


Figure 5.4: ATS Score

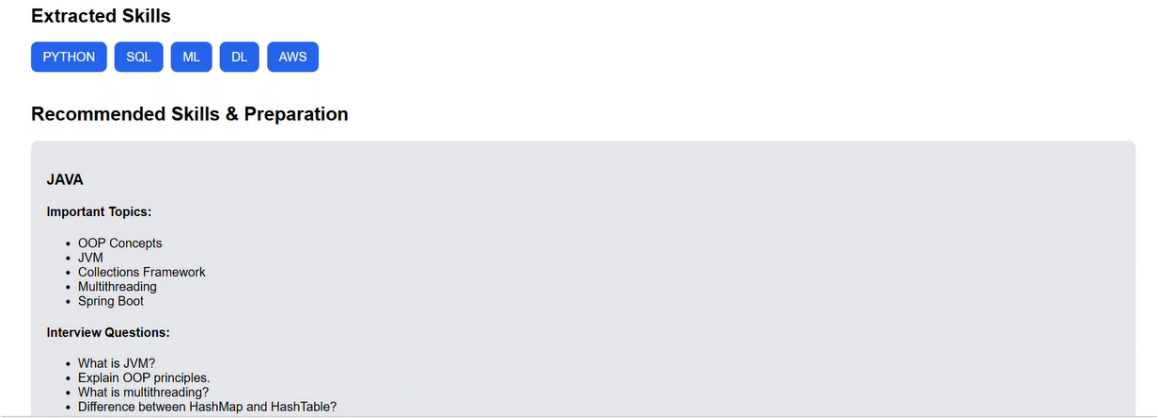


Figure 5.5: Suggested Skills

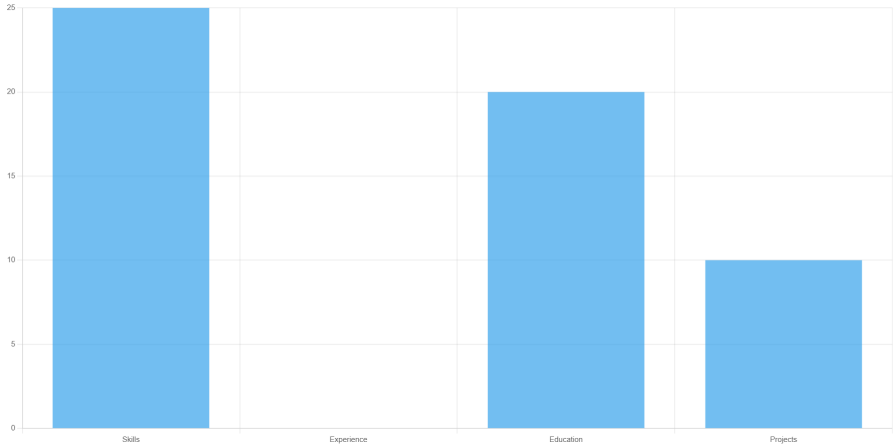


Figure 5.6: Scoring Graph

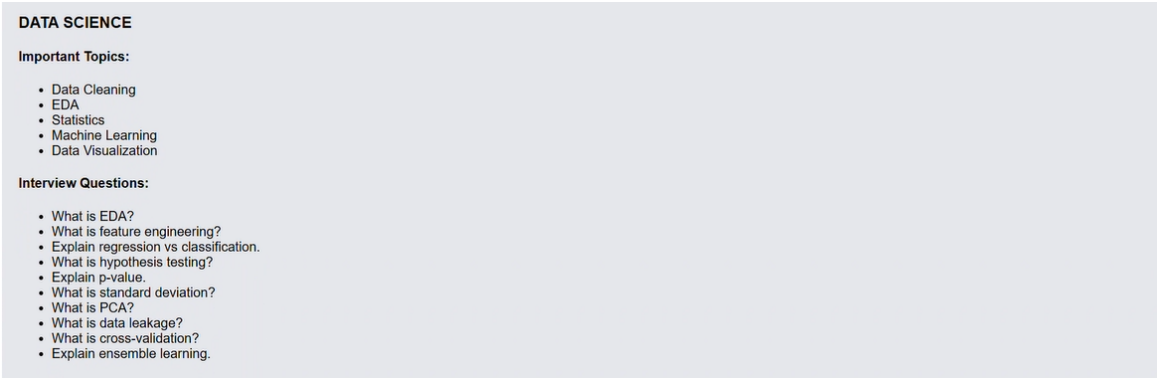
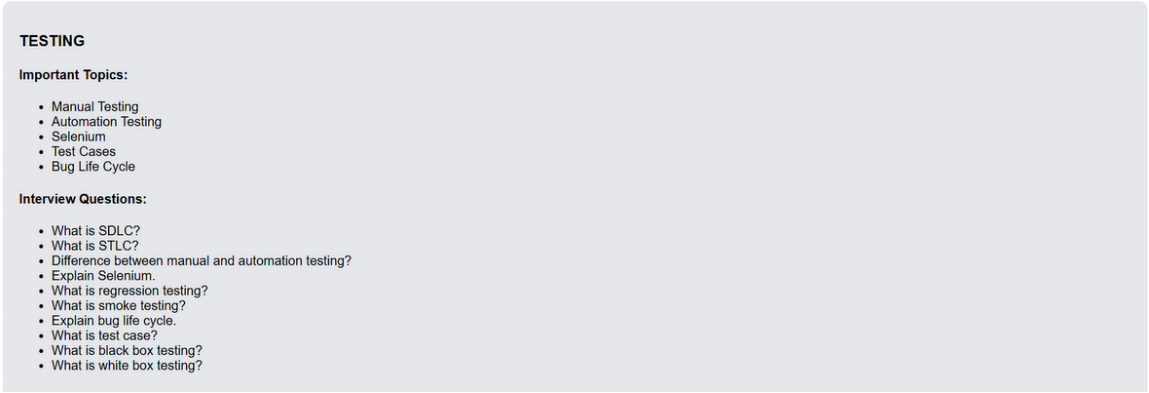


Figure 5.7: Mock Interview Questions



TESTING

Important Topics:

- Manual Testing
- Automation Testing
- Selenium
- Test Cases
- Bug Life Cycle

Interview Questions:

- What is SDLC?
- What is STLC?
- Difference between manual and automation testing?
- Explain Selenium.
- What is regression testing?
- What is smoke testing?
- Explain bug life cycle.
- What is test case?
- What is black box testing?
- What is white box testing?

Figure 5.8: Mock Interview Questions



MINNAKANTI REVATHI

ASPIRING IT PROFESSIONAL

9390947790 | Guntur, Andhra Pradesh, India | m.revathi9390@gmail.com

ABOUT ME

I'm an aspiring IT professional seeking a career in a dynamic, collaborative organization where I can use my technical knowledge and passion for technology. I'm committed to continuous professional development and ready to support innovative projects that boost IT service development.

EDUCATION

2022 - 2026
R.V.R & J.C college of Engineering,Guntur — Bachelor of Technology,ECE, GPA: 8.09

2020 - 2022
Vignan Vidhyalayam Junior College,Guntur — Board of Intermediate Education(MPC), Aggregate: 96%

2019 - 2020
Zilla Parishad High School, Medikonduru — Board of Secondary Education, Aggregate: 97%

PROJECTS

Medical Data Extraction - Python Skill Dzire(2024)
- Developed a Python program to extract, clean, and organize medical records using preprocessing techniques like pattern matching, filtering, and file handling for accuracy.
- Built a Python program to extract and organize medical records into structured data.
- Learned Python basics, data cleaning, file handling, and problem-solving with healthcare data.

Healthcare Management System using SQL
- Developed a healthcare system using SQL to manage patient records, appointments, and billing. Implemented CRUD operations with optimized queries for secure and efficient data management represented data by using Power BI.
- Improved skills in SQL Server integration, data validation, and problem-solving.

CERTIFICATIONS

- Introduction to internet of things-NPTEL-2025
- Sensors and actuators -NPTEL-2025
- Volunteer certificate from NAREDCO-2024

SKILLS

- Python
- DSA(Basics)
- SQL
- OOPS
- Leadership
- Teamwork
- Problem solving
- Communication

Figure 5.9: Resume 2

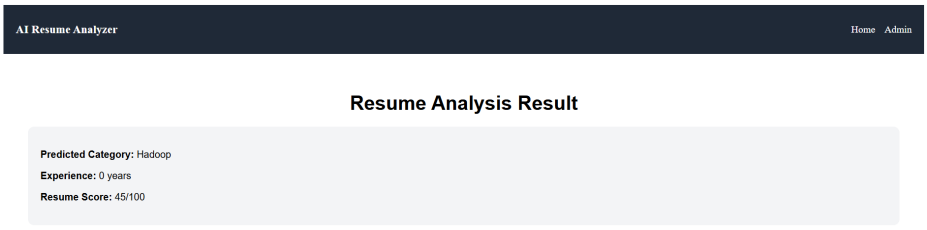


Figure 5.10: ATS Score

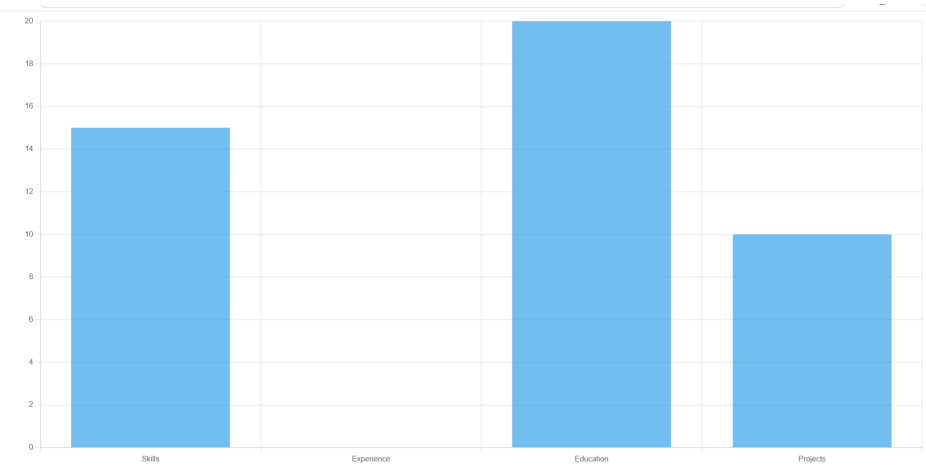


Figure 5.11: Scoring Graph

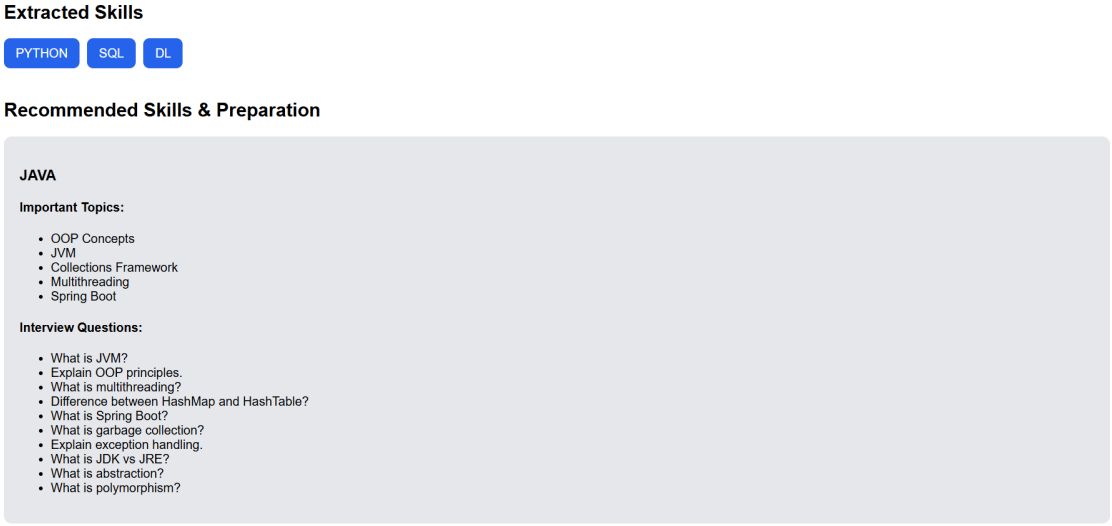
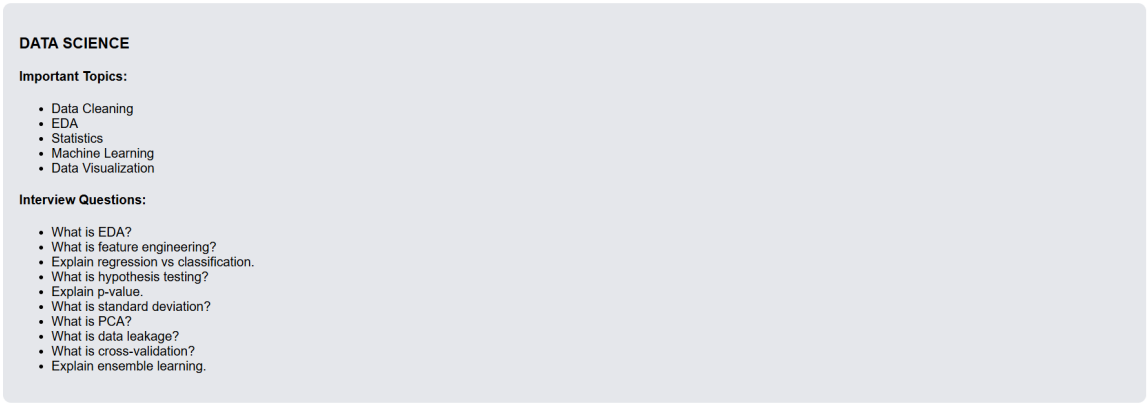


Figure 5.12: Suggetsed Skills



DATA SCIENCE

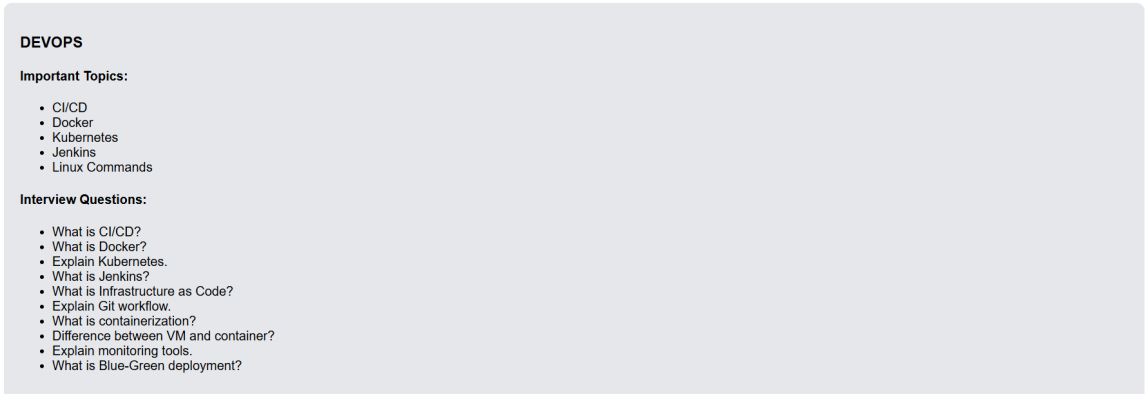
Important Topics:

- Data Cleaning
- EDA
- Statistics
- Machine Learning
- Data Visualization

Interview Questions:

- What is EDA?
- What is feature engineering?
- Explain regression vs classification.
- What is hypothesis testing?
- Explain p-value.
- What is standard deviation?
- What is PCA?
- What is data leakage?
- What is cross-validation?
- Explain ensemble learning.

Figure 5.13: Mock Interview Questions



DEVOPS

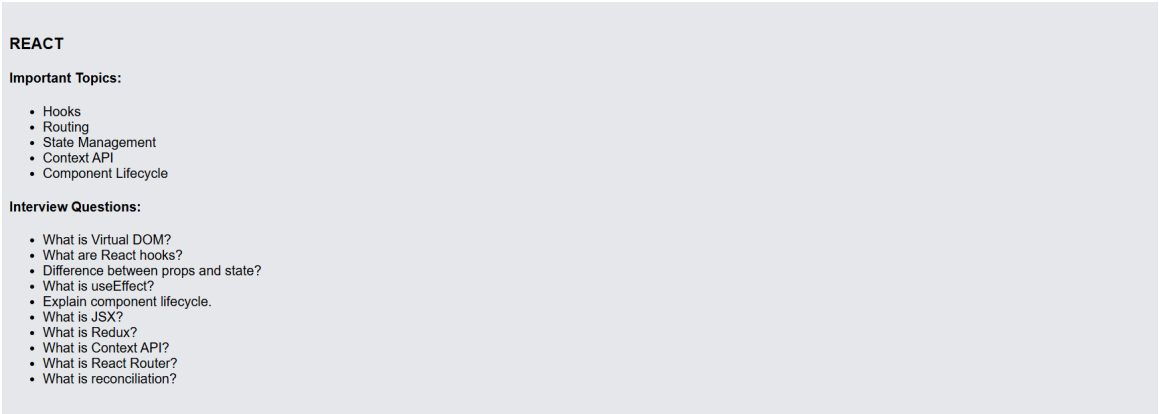
Important Topics:

- CI/CD
- Docker
- Kubernetes
- Jenkins
- Linux Commands

Interview Questions:

- What is CI/CD?
- What is Docker?
- Explain Kubernetes.
- What is Jenkins?
- What is Infrastructure as Code?
- Explain Git workflow.
- What is containerization?
- Difference between VM and container?
- Explain monitoring tools.
- What is Blue-Green deployment?

Figure 5.14: Mock Interview Questions



REACT

Important Topics:

- Hooks
- Routing
- State Management
- Context API
- Component Lifecycle

Interview Questions:

- What is Virtual DOM?
- What are React hooks?
- Difference between props and state?
- What is useEffect?
- Explain component lifecycle.
- What is JSX?
- What is Redux?
- What is Context API?
- What is React Router?
- What is reconciliation?

Figure 5.15: Mock Interview Questions

PONNURU YASASWINI

ponnuruyasaswini5@gmail.com | +91 8688221718
LinkedIn | Github

OBJECTIVE

To approach challenges with a problem-solving mindset by applying skills in Python, Java and SQL, while exploring cloud technologies. I aim to bring adaptability, curiosity, and creativity into every project and contribute to impactful software solutions.

EDUCATION

Vasireddy Venkatadri Institute of Technology — Namburu
B.Tech Computer Science and Engineering (AI&ML) - CGPA - 8.96 — *2022 - 2026*

Sri Chaitanya College — Vijayawada
Intermediate - MPC - *91.1%* — *2020 - 2022*

Sri Chaitanya School — Mangalagiri
SSC - *99.8%* — *2019 - 2020*

SKILLS

Programming :	Python, Java, SQL
Development:	HTML5, CSS3, AWS (EC2, S3, IAM)
Concepts :	Object-Oriented Programming (OOP), Networking
Soft Skills :	Problem Solving, Analytical Thinking, Adaptability, Teamwork

PROJECTS

Smart Resume Match
- Built an AI-powered platform for intelligent resume–job role matching.
- Designed secure workflows for storing and evaluating candidate information.
- Implemented efficient data querying and analysis using Python and SQL.

Full Stack AI SaaS App
- Developed a full-stack AI SaaS product with end-to-end workflow automation.
- Integrated role-based access and cloud resources (AWS) to ensure reliable operations.

CERTIFICATIONS

- Programming in Java - **NPTEL**
- Python for Data Science - **IBM**
- Cloud Fundamentals, Academy Cloud Architecting **- AWS**
- Deep Learning for Computer Vision – **WQU**

INTERNSHIPS

Google Android Developer Virtual Internship | AICTE Eduskills — *April 2024 - June 2024*
Gained hands-on experience in Android app development, focusing on UI/UX design, activity lifecycle, and responsive layouts. Developed and tested Android apps using Android Studio, applying best coding and debugging practices.

ACHIEVEMENTS

- Served as a volunteer at **Make a Difference** for one year
- Secured third place in the **Designathon**, a 24-hour UI/UX **hackathon** held at VVIT, Guntur.

INTERESTS

- Listening to music, Book Reading, Dancing.

Figure 5.16: Resume 3

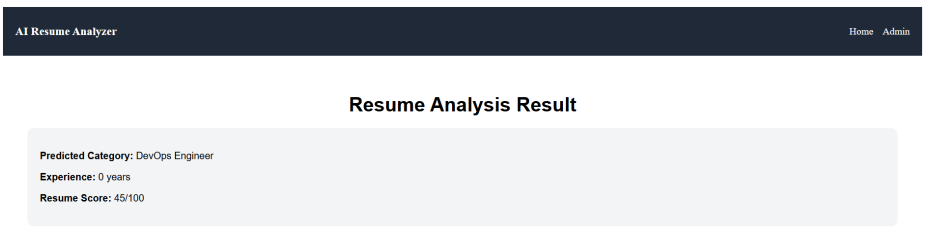


Figure 5.17: ATS Score

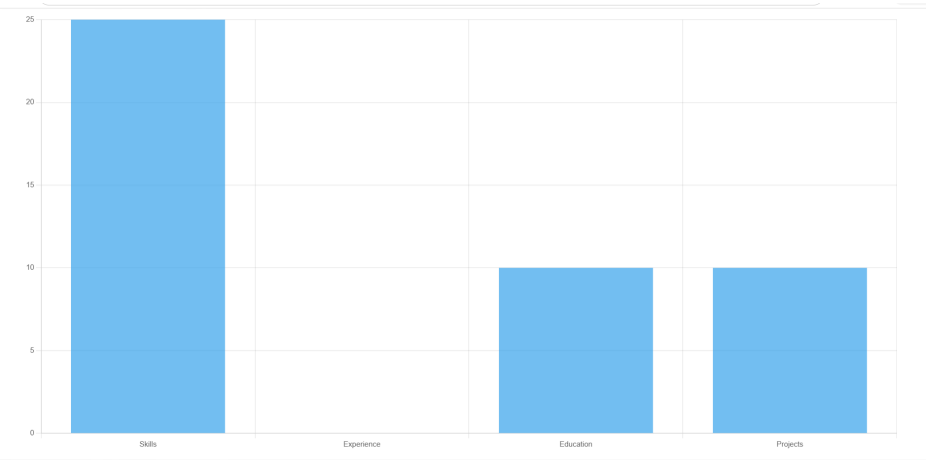


Figure 5.18: Scoring Graph

Extracted Skills

PYTHON JAVA SQL ML AWS

Recommended Skills & Preparation

DATA SCIENCE

Important Topics:

- Data Cleaning
- EDA
- Statistics
- Machine Learning
- Data Visualization

Interview Questions:

- What is EDA?
- What is feature engineering?
- Explain regression vs classification.
- What is hypothesis testing?
- Explain p-value.
- What is standard deviation?
- What is PCA?
- What is data leakage?
- What is cross-validation?
- Explain ensemble learning.

Figure 5.19: Suggetsed Skills

DEVOPS

Important Topics:

- CI/CD
- Docker
- Kubernetes
- Jenkins
- Linux Commands

Interview Questions:

- What is CI/CD?
- What is Docker?
- Explain Kubernetes.
- What is Jenkins?
- What is Infrastructure as Code?
- Explain Git workflow.
- What is containerization?
- Difference between VM and container?
- Explain monitoring tools.
- What is Blue-Green deployment?

Figure 5.20: Mock Interview Questions

TESTING

Important Topics:

- Manual Testing
- Automation Testing
- Selenium
- Test Cases
- Bug Life Cycle

Interview Questions:

- What is SDLC?
- What is STLC?
- Difference between manual and automation testing?
- Explain Selenium.
- What is regression testing?
- What is smoke testing?
- Explain bug life cycle.
- What is test case?
- What is black box testing?
- What is white box testing?

Figure 5.21: Mock Interview Questions

6

CONCLUSION

This study has created and tested an intelligent AI-based system for automatically analyzing resumes and recommending candidates in order to make hiring more efficient. The system effectively analyzes unstructured resume data and generates meaningful insights by combining NLP-based preprocessing, NER-driven information extraction, ATS-basedscoring, and RAG-based recommendation modules.

The analysis of the proposed framework leads to the fol lowing key observations: Using NLP and NER techniques, the system successfully pulls out useful information like skills, education, and experience, which makes sure that candidate profiles arecorrect. The ATS-based scoring system is a reliable and organized way to rank candidates based on how well they fit the job. The combination of RAG makes it possible to give personalized feedback, analyze skill gaps, and makere commendations that are specific to the field, which helps candidates get ready. The system shows that it is more efficient by cutting down on manual work and speeding up decision-makingin hiring processes by using data. The modular architecture ensures scalability and adaptability across multiple domains such as Data Science,DevOps, Testing, and Web Development. The proposed system is a smart and effective way to deal with modern resume screening problems. It makes both the recruiter's job easier and the candidate's experience better

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8

PUBLICATIONS

List of Publications

- Katla Lakshmi Prasanna(Y22EC077),Minnakanti Revathi((Y22EC109), Lakkam Abhilash(Y22EC097), Kotte shasi vardhan Satakarni(Y21EC086), ”RESUME ANALYSIS USING NLP MODLES”, International Journal of Multidisciplinary Research in Science Engineering and Technology on Recent Engineering and Technology (IJMRSET). DOI:10.15680/IJMRSET.2026.0904034 ISSN: 2582-7219, Volume 9, Issue 4, April 2026

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Resume Analysis using NLP Models

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ABSTRACT: Recruiters have a hard time quickly finding good candidates because manually screening resumes takes a lot of time, is not always accurate, and is subject to bias. Current Applicant Tracking Systems (ATS) depend too much on keyword matching and don't do a good job of capturing semantic relevance and understanding context. This paper proposes an AI-based resume analysis and Recommendation System that combines Natural Language Processing (NLP), Named Entity Recognition (NER), Applicant Tracking System (ATS) scoring, and Retrieval-Augmented Generation (RAG) to overcome these limitations. The suggested system automates the parsing of resumes, pulls out important information like skills, education, and experience, and decides if a candidate is a good fit for the job based on the requirements. It also gives personalized feedback, analyzes skill gaps, and makes up mock interview questions to help candidates get ready. The system makes hiring easier, cuts down on bias, and lets you make decisions based on data. The results of the experiment show that this method is more accurate and better at matching meanings than traditional methods.

KEYWORDS: Applicant Tracking Systems, Natural Language Processing, Named Entity Recognition, Retrieval-Augmented Generation.

I. INTRODUCTION

Hiring is an important part of running a business today, but most of the time, traditional methods for screening resumes are still manual, take a long time, and aren't always reliable. Recruiters often look at a lot of resumes in a short amount of time, which makes it more likely that they will miss qualified candidates. Current Applicant Tracking Systems (ATS) mainly use keyword-based filtering, which doesn't work well for understanding the meaning and context of resumes. Because of this, candidates who might be a good fit could be turned down because their keywords don't match well, which makes the hiring process less efficient and more biased.

With AI and NLP making quick progress, automated resume analysis has become a promising way to deal with these problems. This paper presents an intelligent resume analysis and recommendation system that incorporates NLP techniques, Named Entity Recognition (NER), ATS-based scoring, and Retrieval-Augmented Generation (RAG). The system not only automatically parses resumes and rates candidates, but it also gives personalized feedback, analyses skill gaps, and creates mock interview questions. This method makes hiring more efficient, helps people make better decisions, and helps candidates get ready for job opportunities.

II. LITERATURE SURVEY

A. Traditional Resume Screening Methods

The main ways that traditional resume screening works are by hand and with simple keyword-based filtering. Recruiters look at resumes by matching certain keywords that are related to job descriptions, skills, and qualifications. This method is easy to understand, but it takes a lot of time, isn't always accurate, and is open to human bias. Also, keyword-based filtering doesn't take into account the meaning of the words, which means that it might turn down candidates who could be a good fit but don't use exact keyword matches.

B. AI-Based Resume Screening Systems

Recent improvements in Artificial Intelligence (AI) have made it possible to create systems that automatically screen resumes. The conference paper talks about an intelligent resume screening tool that uses Natural Language Processing

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(NLP) and machine learning to find useful information in resumes and sort candidates by job role. These systems make screening faster and easier by cutting down on the amount of work that needs to be done by hand. But most AI-based systems still depend a lot on pattern matching and don't really.

C. NLP and Named Entity Recognition Techniques

Tokenization, stemming, and lemmatization are all examples of Natural Language Processing (NLP) techniques that are commonly used to process resume data. Named Entity Recognition (NER) is very important for getting structured information like a candidate's name, skills, education, and experience from text that isn't structured. These methods make it easier to get accurate data and better candidate profiles. Even with these improvements, many systems still don't do a good job of combining semantic analysis with decision making.

D. Limitations of Existing Methods

Even though automated resume analysis has come a long way, there are still some problems. Traditional Applicant Tracking Systems (ATS) don't understand semantics and mostly rely on matching keywords. Many AI-based solutions don't give personalized feedback, analyze skill gaps, or help people get ready for interviews. Also, their decisions are not clear or easy to understand. The proposed system combines NLP, NER, ATS scoring, and Retrieval-Augmented Generation (RAG) to make a more complete and smart solution to these problems.

III. PROPOSED FRAMEWORK

The proposed system is a smart pipeline that will automatically analyze resumes and suggest candidates. It uses Natural Language Processing (NLP), Named Entity Recognition (NER), Applicant Tracking System (ATS) scoring, and Retrieval-Augmented Generation (RAG) to quickly and accurately evaluate resumes and give useful information. The system processes resumes in a set order, turning unstructured text into useful information.

A. System Overview

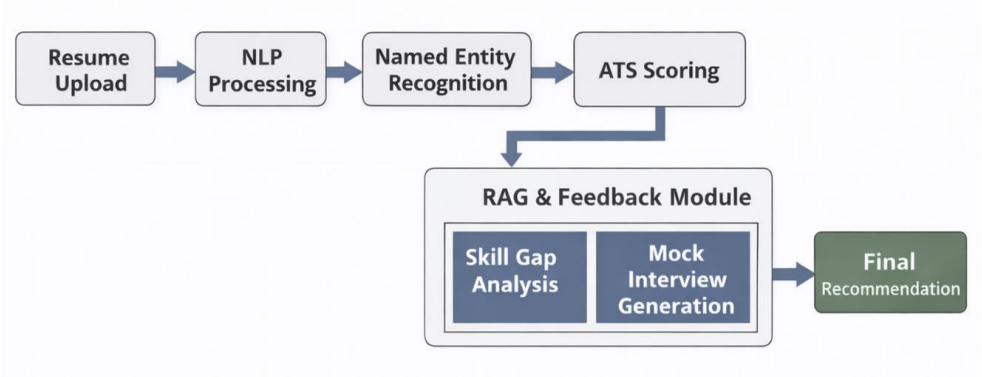


FIG 1: Proposed Framework.

Figure 1 shows the overall design of the proposed system. There are many parts to the system, such as resume input, preprocessing, information extraction, evaluation, and feedback generation. To make sure that resume analysis is accurate and quick, each module does a different job. The pipeline makes sure that data flows smoothly from the raw input to the final output.

B. Resume Input and Preprocessing

{Step 1}: Resume Upload

The system can read resumes in different formats, like PDF and DOCX. These resumes may have unstructured text data that is laid out and styled in different ways.

{Step 2}: Text Extraction

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Document parsing techniques turn the uploaded resume into plain text. This step makes sure that the content can be processed further.

{Step 3}: Data Preprocessing

The text that was taken out goes through preprocessing steps like tokenization, removing stop words, stemming, and lemmatization. These steps help clean and normalize the data, which makes downstream NLP tasks work better.

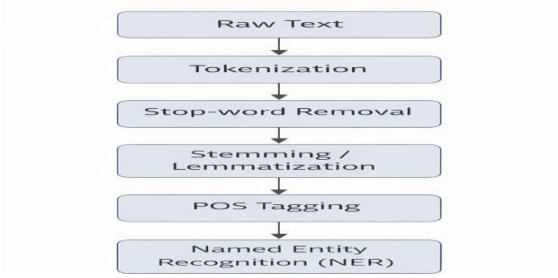


FIG 2 : NLP Preprocessing and Feature Extraction Process.

Figure 2 shows the steps that were taken to turn raw resume text into structured features that can be used for more analysis.

C. NLP Processing and Feature Extraction

{Step 4}: Linguistic Processing

We use Natural Language Processing to figure out what the resume text means and how it is put together. This includes tagging parts of speech and breaking sentences down.

{Step 5}: Named Entity Recognition (NER)

NER is used to get important information about candidates, like their name, skills, education, work history, and certifications. These entities are organized and kept for later review. The extracted features serve as the foundation for candidate evaluation

D. ATS Scoring Mechanism.

The ATS scoring mechanism and model structure are shown in Fig. 3.

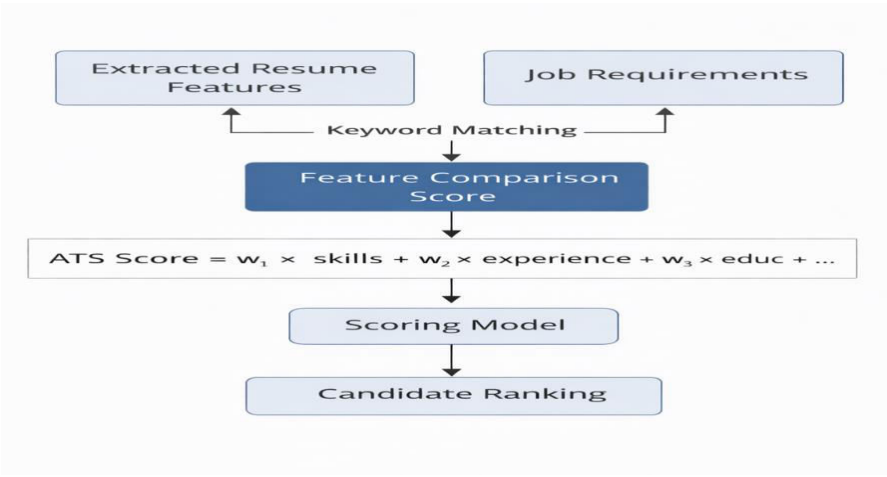


FIG 3. ATS Scoring and Ranking flowchart

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{Step 6}: Job Description Matching
The system compares the extracted resume features to the job requirements using keyword matching and semantic similarity techniques.
{Step 7}: Scoring and Ranking
An ATS-based scoring system gives different weights to things like skills, experience, and education. Candidates are ranked based on how well they fit the job description. This helps recruiters find the best candidates quickly.

E. Retrieval-Augmented Generation (RAG) Module
{Step 8}: Contextual Information Retrieval
The RAG module gets useful information from outside sources of knowledge to help people make better decisions.
{Step 9}: Feedback and Recommendation Generation
The system creates personalized feedback based on the analysis, which includes an analysis of skill gaps and suggestions for how to improve. This gives candidates more than just a simple screening.

F. Mock Interview Generation
{Step 10}: Question Generation
Using Natural Language Generation techniques, the system makes interview questions that are specific to each role.
{Step 11}: Candidate Preparation Support
The questions that are made help candidates get ready for interviews. You can also use Text-to-Speech (TTS) modules to give interactive help.

G. System Workflow
Figure 4 shows how the proposed system works from start to finish. The process starts with uploading a resume and then goes through preprocessing, entity extraction, scoring, and generating feedback. The final product includes a list of candidates ranked, a detailed analysis, and help getting ready for interviews.

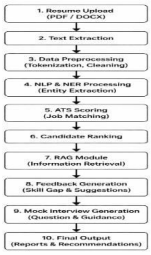


FIG 4: Work Flow

H. Performance Monitoring
The system's performance is judged by how well it extracts entities, how relevant job matches are, and how useful recommendations are. The model makes sure that NLP processing and scoring work well by optimizing them.

IV. RESULTS AND ANALYSIS

The suggested AI-based resume analysis system is tested by putting it into action and showing the results. The system shows that it works by processing resumes, getting useful information from them, and making smart suggestions.

A. Input Resume
The system can read resumes that are in PDF or DOCX format. Figure 5 shows an example of a resume that was sent to the system. The input has unstructured text that includes personal information, skills, education, and work history

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SUMMARY

I am an experienced Data Engineer skilled in PySpark, Google Cloud Platform(GCP), Python, SQL and Airflow. I specialize in building and optimizing data pipelines. I constantly commit to continuous learning and leveraging new technologies to enhance data solutions.

PROFESSIONAL EXPERIENCE

Data Engineer 07/2021 – present
Tata Consultancy Services Hyderabad, India

- Build ETL/ELT/EL batch data pipelines to move data from on-premise to Data Warehouse (**BigQuery**) and vice-versa using **Pyspark**, **SparkSQL** and **Apache Airflow**
- Iteratively design and develop a modular, reusable PySpark and Airflow based **framework** to create and orchestrate distributed data processing pipelines
- Fine-tuned and migrated oracle **stored procedures** and **packages** to Bigquery
- Optimize and performance tune big data pipelines to handle TBs of data using Apache Spark and SQL
- Perform proof of concepts(**POC**) using interactive development environments such as Jupyter Notebooks
- Debug erroneous data pipelines including undocumented legacy pipelines and troubleshoot complex data issues and provide solutions
- Create **Bash** or **Python** scripts to **automate** routine tasks

EDUCATION

V R SIDDHARTHA ENGINEERING COLLEGE 2017 – 2021
Bachelor of Technology in Electronics and Communication Engineering Vijayawada, India
CGPA: 8.61

PROJECTS

Smart AI Doc Miner
Generative AI, Flask, Python, Google Cloud Platform, pgvector, Google Gemini 1.5

- Implemented **RAG (Retrieval-Augmented Generation)** methodology to enable efficient document retrieval and analysis.
- Integrated **Google Gemini 1.5 model** for state-of-the-art natural language processing and understanding.
- Utilized **pgvector** for vector storage and similarity search, enhancing data retrieval capabilities.
- **Deployment**: Hosted the application on **Google Cloud Run**, ensuring scalability and seamless deployment.

SKILLS

Programming Languages — Python, SQL, Shell Scripting

Frameworks — Apache Spark (Pyspark), Apache Airflow

Cloud Services — **Google cloud platform (GCP)** - BigQuery, Google Cloud Storage(GCS), Dataproc, Pub/Sub, Cloud Composer, Google Compute Engine, Google Kubernetes Engine, IAM, Cloud SQL, Cloud Functions, Vertex AI, Cloud Run, Cloud Shell, Cloud Scheduler, Artifact Registry | **Amazon Web Services (AWS)** - S3, Lambda, DynamoDB | **Microsoft Azure** - Azure Data Factory, Azure Databricks, Azure Synapse Analytics

ETL Tools — IBM InfoSphere DataStage

Orchestration Tools — Cloud Composer, Control-M

DBMS — Teradata, postgres, oracle

CERTIFICATES

- Google Cloud Certified Professional Data Engineer
- Google Cloud Certified Associate Data Practitioner
- Google Cloud GenAI L400 Certified

AWARDS

Employee of the Month x 3

- On three different occasions, I have been awarded Employee of the Month for my individual contributions

Project Performance Improvement Award x2

- As part of a 5 member team, helped the customer save more than $150k
- As part of a 2 member team, worked to meet Government's Data Regulatory policies within a strict deadline of 4 months

INTERESTS

Cricket | Volleyball | Badminton | Reading | Photography

FIG 5: Resume input

B. User Interface for Resume Upload



FIG 6: Resume Upload Interface

A user-friendly front-end interface lets users upload resumes and start the analysis process. The system can read files in PDF and DOCX formats, so you can use a variety of input types. Once the resume is uploaded, it is processed automatically without anyone having to do anything. Figure 6 shows the screen where users choose and upload resumes for analysis. The design makes it easy to get around and find results quickly, which improves the overall user experience.

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C. Resume Analysis Result

After processing, the system makes a summary of the candidate's profile. The output in Fig. 7 shows the predicted job category, level of experience, and overall resume score. The scoring system based on ATS looks at the resume and gives a numerical score.



FIG 7: Resume Analysis Result

D. Feature-Based Performance Analysis

The system looks at different factors, like skills, education, work experience, and projects. Figure 8 shows these features in a graph, showing how they affect the overall score. This helps you figure out what parts of the resume are good and what parts need work.

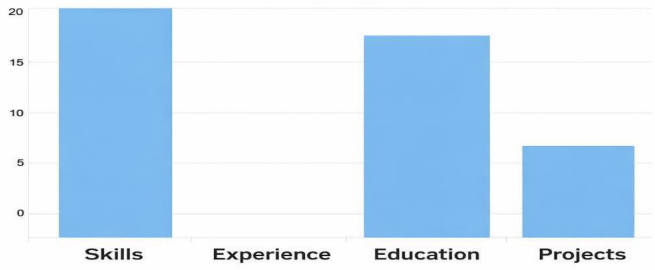


FIG 8: Resume Analysis Result

E. Extracted Skills and Domain-Specific Recommendations

By utilizing NLP and NER modules, the system extracts relevant skills from resumes to identify deficiencies and provide tailored preparation guidelines as illustrated in Figure 9. Beyond basic skill assessment, the system generates domain-specific recommendations, such as the Data Science interview topics and questions shown in Figure 10, to help candidates meet industry standards.

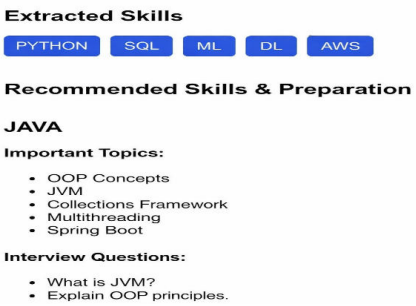


FIG 9: Extracted Skills and Recommendations



FIG 10: Data Science Recommendations

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F. DevOps Domain Analysis

In the same way, Fig. 11 shows suggestions and interview questions for the DevOps field. The system changes its suggestions based on the skills it sees and the job role it thinks the person will have.



FIG 11: DevOps Recommendations

V. CONCLUSION

This study has created and tested an intelligent AI-based system for automatically analyzing resumes and recommending candidates in order to make hiring more efficient. The system effectively analyzes unstructured resume data and generates meaningful insights by combining NLP-based preprocessing, NER-driven information extraction, ATS-based scoring, and RAG-based recommendation modules.

The analysis of the proposed framework leads to the following key observations:

1. Using NLP and NER techniques, the system successfully pulls out useful information like skills, education, and experience, which makes sure that candidate profiles are correct.
2. The ATS-based scoring system is a reliable and organized way to rank candidates based on how well they fit the job.
3. The combination of RAG makes it possible to give personalized feedback, analyze skill gaps, and make recommendations that are specific to the field, which helps candidates get ready.
4. The system shows that it is more efficient by cutting down on manual work and speeding up decision-making in hiring processes by using data.
5. The modular architecture ensures scalability and adaptability across multiple domains such as Data Science, DevOps, Testing, and Web Development.

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